

Vishwavrinda Ayurveda API And System Documentation

This document describes the backend software, API contract, authentication, data flow, and important file locations for the Vishwavrinda Ayurveda WhatsApp AI admin system.

Application Layout

```
auryvedic/  
backend/  
src/  
constants/ API paths, auth constants, pagination, database names, indexes  
lib/ shared helpers for pagination, ObjectId, regex, body validation  
parsers/ WhatsApp webhook payload parsing  
plugins/ Fastify plugins such as JWT authentication  
routes/ HTTP route handlers  
services/ MongoDB, OpenAI, WhatsApp, appointment, dashboard, knowledge logic  
types/ TypeScript models and Fastify/JWT augmentation  
docs/ API and system documentation  
scripts/ demo seed and PDF generation scripts  
knowledge.md file-based clinic knowledge used by the AI  
frontend/  
app/ Next.js dashboard pages  
components/ shadcn-style UI components  
lib/ API client, auth helpers, formatting  
proxy.ts frontend route protection
```

Authentication

Admin dashboard APIs are protected with JWT.

Public backend routes:

- 'GET /'
- 'GET /webhook'
- 'POST /webhook'
- 'POST /api/auth/login'
- 'OPTIONS *'

Dev-only protected backend routes, registered only when ENABLE_TEST_ENDPOINTS=true:

- 'GET /test-whatsapp'
- 'GET /test-ai'

Protected backend routes:

- 'POST /api/ai/test'
- 'GET /api/dashboard/stats'
- 'GET /api/appointments'
- 'PATCH /api/appointments/:id'
- 'GET /api/users'
- 'GET /api/users/:phoneNumber/messages'

- 'GET /api/knowledge'
- 'PUT /api/knowledge'

Admin credentials are read from environment variables:

```
ADMIN_USERNAME
ADMIN_PASSWORD
role: admin
```

Required backend environment variables:

```
VERIFY_TOKEN
JWT_SECRET
ADMIN_USERNAME
ADMIN_PASSWORD
OPENAI_API_KEY
WHATSAPP_TOKEN
PHONE_NUMBER_ID
```

Authenticated requests must include:

```
Authorization: Bearer <jwt>
```

Relevant files:

- 'backend/src/plugins/auth.ts'
- 'backend/src/routes/auth.ts'
- 'backend/src/constants/auth.ts'
- 'backend/src/types/auth.ts'
- 'frontend/app/login/page.tsx'
- 'frontend/proxy.ts'
- 'frontend/lib/api-client.ts'

System Flow

```
Patient on WhatsApp
|
v
Meta WhatsApp Cloud API
|
v
POST /webhook
|
v
backend/src/parsers/whatsapp.ts
|
v
backend/src/services/conversation.ts
|
+--> upsertUser(phoneNumber)
| |
| v
| MongoDB users
|
+--> createMessage(role=user)
| |
| v
```

```

| MongoDB messages
|
+--> getKnowledge()
| |
| v
| backend/knowledge.md
|
+--> generateReply()
| |
| v
| OpenAI
|
+--> AIResult type=message
| |
| v
| sendAndStoreMessage()
| |
| +--> sendMessages() to WhatsApp
| |
| +--> createMessage(role=assistant) only after successful send
|
+--> AIResult type=appointment
|
v
createAppointment()
|
v
MongoDB appointments

```

Admin Dashboard Flow

```

Admin Browser
|
v
frontend/proxy.ts checks JWT cookie
|
+--> no valid token -> /login
|
+--> valid admin token -> dashboard page
|
v
frontend/lib/api-client.ts adds Authorization header
|
v
Fastify preHandler verifies JWT and role=admin
|
v
backend route handler

```

Data Stores

MongoDB collections:

- 'users'
- 'messages'
- 'appointments'

File storage:

- 'backend/knowledge.md'

Indexes created on startup:

- 'users.phoneNumber' unique
- 'users.lastActiveAt'
- 'users.firstSeenAt'
- 'messages.phoneNumber + createdAt'
- 'appointments.createdAt'
- 'appointments.isCompleted + createdAt'
- 'appointments.phoneNumber'

Common Pagination

Query parameters:

- 'page': positive integer, default '1'
- 'limit': positive integer, default '20', maximum '100'

Response shape:

```
{
  "data": [],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 0,
    "totalPages": 0
  }
}
```

API Reference

GET /

File:

- 'backend/src/index.ts'

Auth:

- Public

Purpose:

- Health check endpoint.

Parameters:

- None

Returns:

```
{
  "status": "ok",
  "service": "WhatsApp AI Bot"
}
```

POST /api/auth/login

Files:

- 'backend/src/routes/auth.ts'
- 'backend/src/plugins/auth.ts'

Auth:

- Public

Purpose:

- Login for the single hardcoded admin user.

Body:

```
{
  "username": "admin",
  "password": "<admin password>"
}
```

Returns:

```
{
  "token": "jwt-token",
  "user": {
    "username": "admin",
    "role": "admin"
  }
}
```

Errors:

- '401' when username or password is incorrect.

GET /webhook

Files:

- 'backend/src/routes/webhook.ts'
- 'backend/src/config.ts'

Auth:

- Public

Purpose:

- Meta WhatsApp webhook verification.

Query parameters:

- 'hub.mode': expected 'subscribe'
- 'hub.verify_token': must match 'VERIFY_TOKEN'
- 'hub.challenge': challenge string returned to Meta

Returns:

- '200' with challenge string when valid.
- '403' with 'Verification failed' when invalid.

POST /webhook

Files:

- 'backend/src/routes/webhook.ts'
- 'backend/src/parsers/whatsapp.ts'
- 'backend/src/services/conversation.ts'

Auth:

- Public

Purpose:

- Receives WhatsApp events, stores messages, calls the AI, sends replies, and creates appointments when the AI returns complete appointment data.

Body:

- Meta WhatsApp webhook payload.

Returns:

```
{
  "success": true
}
```

Side effects:

- Upserts user in 'users'.
- Stores incoming user message in 'messages'.
- Reads 'backend/knowledge.md'.
- Calls OpenAI.
- Sends WhatsApp message.
- Stores assistant reply only after WhatsApp send succeeds.
- Creates appointment in 'appointments' when needed.

GET /test-whatsapp

File:

- 'backend/src/routes/webhook.ts'

Auth:

- Admin JWT required

Purpose:

- Sends a fixed WhatsApp test message.

Parameters:

- None

Returns:

```
{
  "success": true
}
```

GET /test-ai

Files:

- 'backend/src/routes/webhook.ts'
- 'backend/src/services/knowledge.ts'
- 'backend/src/services/openai.ts'

Auth:

- Admin JWT required

Purpose:

- Loads knowledge and sends a fixed test question to OpenAI.

Parameters:

- None

Returns:

```
{
  "success": true,
  "reply": {
    "type": "message",
    "reply": "string",
    "appointment": null
  }
}
```

POST /api/ai/test

Files:

- 'backend/src/routes/ai.ts'
- 'backend/src/services/knowledge.ts'
- 'backend/src/services/openai.ts'

Auth:

- Admin JWT required

Purpose:

- Tests the AI response flow from an admin/API client without sending a WhatsApp message.
- Loads 'backend/knowledge.md', calls OpenAI, and returns the structured AI result.
- Does not create users, messages, or appointments in MongoDB.

Body:

```
{
  "message": "What is the consultation fee?"
}
```

Returns:

```
{
  "success": true,
  "result": {
    "type": "message",
    "reply": "The consultation fee is Rs. 350...",
    "appointment": null
  }
}
```

Appointment result shape:

```
{
  "success": true,
  "result": {
    "type": "appointment",
    "reply": "Thanks. The clinic team will contact you soon.",
    "appointment": {
      "patientName": "Raj Sharma",
      "preferredDate": "2026-08-15",
      "preferredTime": "5 PM",
      "preferredContactMethod": "whatsapp",
      "reason": "Acidity"
    }
  }
}
```

Errors:

- '400' when message is missing or empty.
- '401' when JWT is missing or invalid.
- '502' when the AI request fails.

GET /api/dashboard/stats

Files:

- 'backend/src/routes/dashboard.ts'
- 'backend/src/services/dashboard.ts'

Auth:

- Admin JWT required

Purpose:

- Returns counters and chart trend data for the dashboard.

Query parameters:

- 'period': optional 'week' or 'month', default 'week'
- 'anchorDate': optional date in 'YYYY-MM-DD'; when omitted, today is used

Returns:

```
{
  "totalUsers": 10,
  "newUsersThisWeek": 4,
  "newUsersThisMonth": 8,
  "totalMessages": 40,
  "totalAppointments": 10,
  "pendingAppointments": 6,
  "completedAppointments": 4,
  "appointmentsThisWeek": 3,
  "appointmentsThisMonth": 9,
  "trend": {
    "period": "week",
    "anchorDate": "2026-07-09",
    "startDate": "2026-07-03",
    "endDate": "2026-07-09",
    "points": [
      {
        "date": "2026-07-03",
        "label": "3 Jul",
        "users": 5,
        "appointments": 2
      }
    ]
  }
}
```

Errors:

- '400' for invalid period.
- '400' for invalid anchor date.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

GET /api/appointments

Files:

- 'backend/src/routes/appointments.ts'
- 'backend/src/services/appointments.ts'

Auth:

- Admin JWT required

Purpose:

- Returns paginated appointment requests.

Query parameters:

- 'page': optional positive integer
- 'limit': optional positive integer, max '100'
- 'status': optional 'pending', 'done', or 'all', default 'all'
- 'dateCategory': optional 'today', 'upcoming', 'past', or 'all', default 'all'
- 'search': optional search across 'patientName' and 'phoneNumber'

Returns:

```
{
  "data": [
    {
      "_id": "64f000000000000000000001",
      "patientName": "Aarav Sharma",
      "phoneNumber": "+919880000001",
      "preferredDate": "2026-07-10",
      "preferredTime": "10:30 AM",
      "reason": "Digestive discomfort",
      "preferredContactMethod": "whatsapp",
      "source": "whatsapp",
      "isCompleted": false,
      "createdAt": "2026-07-09T10:00:00.000Z",
      "updatedAt": "2026-07-09T10:00:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 1,
    "totalPages": 1
  }
}
```

Errors:

- '400' for invalid pagination.
- '400' for invalid status.
- '400' for invalid date filter.
- '401' when JWT is missing or invalid.

- '500' for unexpected database errors.

PATCH /api/appointments/:id

Files:

- 'backend/src/routes/appointments.ts'
- 'backend/src/services/appointments.ts'

Auth:

- Admin JWT required

Purpose:

- Updates only appointment completion status.

Path parameters:

- 'id': MongoDB ObjectId string

Body:

```
{
  "isCompleted": true
}
```

Returns:

```
{
  "success": true,
  "data": {
    "_id": "64f000000000000000000001",
    "patientName": "Aarav Sharma",
    "phoneNumber": "+919880000001",
    "preferredDate": "2026-07-10",
    "preferredTime": "10:30 AM",
    "reason": "Digestive discomfort",
    "preferredContactMethod": "whatsapp",
    "source": "whatsapp",
    "isCompleted": true,
    "createdAt": "2026-07-09T10:00:00.000Z",
    "updatedAt": "2026-07-09T10:15:00.000Z",
    "completedAt": "2026-07-09T10:15:00.000Z"
  }
}
```

Rules:

- Only 'isCompleted' can be changed.
- Extra body fields are rejected.
- Setting 'isCompleted=true' adds 'completedAt'.
- Setting 'isCompleted=false' removes 'completedAt'.

Errors:

- '400' for invalid Objectld.
- '400' for invalid body.
- '401' when JWT is missing or invalid.
- '404' when appointment is not found.
- '500' for unexpected database errors.

GET /api/users

Files:

- 'backend/src/routes/users.ts'
- 'backend/src/services/users.ts'

Auth:

- Admin JWT required

Purpose:

- Returns paginated WhatsApp users.

Query parameters:

- 'page': optional positive integer
- 'limit': optional positive integer, max '100'
- 'search': optional phone number search

Returns:

```
{
  "data": [
    {
      "phoneNumber": "+919880000001",
      "firstSeenAt": "2026-07-01T10:00:00.000Z",
      "lastActiveAt": "2026-07-09T10:00:00.000Z",
      "totalMessages": 4,
      "createdAt": "2026-07-01T10:00:00.000Z",
      "updatedAt": "2026-07-09T10:00:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 1,
    "totalPages": 1
  }
}
```

Errors:

- '400' for invalid pagination.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

GET /api/users/:phoneNumber/messages

Files:

- 'backend/src/routes/users.ts'
- 'backend/src/services/messages.ts'

Auth:

- Admin JWT required

Purpose:

- Returns paginated conversation history for one WhatsApp phone number.

Path parameters:

- 'phoneNumber': WhatsApp phone number

Query parameters:

- 'page': optional positive integer
- 'limit': optional positive integer, max '100'

Returns:

```
{
  "data": [
    {
      "phoneNumber": "+919880000001",
      "role": "user",
      "content": "Namaste, I need an appointment.",
      "createdAt": "2026-07-09T10:00:00.000Z"
    },
    {
      "phoneNumber": "+919880000001",
      "role": "assistant",
      "content": "Please share your preferred date and time.",
      "createdAt": "2026-07-09T10:01:00.000Z"
    }
  ],
  "pagination": {
    "page": 1,
    "limit": 20,
    "total": 2,
    "totalPages": 1
  }
}
```

Errors:

- '400' for invalid pagination.
- '400' for invalid phone number.
- '401' when JWT is missing or invalid.
- '500' for unexpected database errors.

GET /api/knowledge

Files:

- 'backend/src/routes/knowledge.ts'
- 'backend/src/services/knowledge.ts'

Auth:

- Admin JWT required

Purpose:

- Reads file-based clinic knowledge.

Parameters:

- None

Returns:

```
{  
  "content": "markdown content"  
}
```

Errors:

- '401' when JWT is missing or invalid.
- '500' for filesystem errors.

PUT /api/knowledge

Files:

- 'backend/src/routes/knowledge.ts'
- 'backend/src/services/knowledge.ts'

Auth:

- Admin JWT required

Purpose:

- Replaces 'backend/knowledge.md'.

Body:

```
{  
  "content": "updated markdown content"  
}
```

Validation:

- 'content' must be a string.

- 'content' must not be empty after trimming.
- 'content' maximum length is '100000' characters.

Returns:

```
{  
  "success": true  
}
```

Errors:

- '400' for invalid content.
- '401' when JWT is missing or invalid.
- '500' for filesystem errors.

Important Files By Responsibility

- Backend entrypoint: 'backend/src/index.ts'
- Route registration: 'backend/src/routes/index.ts'
- Auth plugin: 'backend/src/plugins/auth.ts'
- API path constants: 'backend/src/constants/api.ts'
- Database constants: 'backend/src/constants/database.ts'
- Pagination helper: 'backend/src/lib/pagination.ts'
- Request body validators: 'backend/src/lib/request-body.ts'
- WhatsApp parser: 'backend/src/parsers/whatsapp.ts'
- WhatsApp sender: 'backend/src/services/whatsapp.ts'
- OpenAI service: 'backend/src/services/openai.ts'
- Conversation orchestrator: 'backend/src/services/conversation.ts'
- Appointment service: 'backend/src/services/appointments.ts'
- User service: 'backend/src/services/users.ts'
- Message service: 'backend/src/services/messages.ts'
- Dashboard service: 'backend/src/services/dashboard.ts'
- Knowledge service: 'backend/src/services/knowledge.ts'
- Frontend API client: 'frontend/lib/api-client.ts'
- Frontend login page: 'frontend/app/login/page.tsx'
- Frontend protected route proxy: 'frontend/proxy.ts'

Final Local Test Report

Date:

- 2026-08-11

Result:

- Backend route/security sweep: '26/26' passed.
- Backend typecheck: passed.
- Frontend typecheck: passed with '--incremental false'.

- Frontend lint: passed.
- Backend npm audit: '0 vulnerabilities'.
- Frontend npm audit: '0 vulnerabilities'.
- WhatsApp E2E simulation: passed.

Verified route behavior:

- Public health endpoint returns '200'.
- Webhook verification accepts the correct 'VERIFY_TOKEN' and rejects invalid tokens.
- Login returns a JWT for valid admin credentials and rejects invalid credentials.
- Protected '/api/*' routes reject missing JWT with '401'.
- Validation errors return '400' for bad AI body, bad pagination, bad dashboard filters, bad appointment status/date filter, bad appointment update body, invalid appointment id, and invalid knowledge content.
- AI test answers clinic fee questions from 'knowledge.md'.
- WhatsApp E2E simulation saved the user, stored user and assistant messages, sent WhatsApp replies, created an appointment, and updated dashboard counts.